Supporting Information

S1 Procedure of the algorithm

Algorithm 1 shows the procedure of our algorithm. In each iteration, we update the distribution along with the measurement results. For ε, t, and $\vec{x}$ (i.e., the computational-basis measurement result represented by a binary sequence), there is a relation among prior distribution $P(\varepsilon)$, likelihood function $P(\vec{x}|\varepsilon)$, and posterior distribution $P(\varepsilon|\vec{x})$ called the Bayes theorem,

$$P(\varepsilon|\vec{x}) \propto P(\vec{x}|\varepsilon)P(\varepsilon). \tag{S1.1}$$

The calculated posterior distribution is used as the prior distribution in the next iteration. In this work, we take the Bayes setting with Gaussian probability distribution $P(z) = \mathcal{N}(z; \mu, \sigma^2)$ of the random variable z, with mean μ and variance σ^2. We specifically choose the Gaussian prior $P(\varepsilon) = \mathcal{N}(\varepsilon; \mu_{\text{prior}}, \sigma^2_{\text{prior}})$.

This study considers distributions over two patterns for all qubits $\vec{x} \in \{\vec{0}, \text{others}\}$ where $\vec{0}$ represents all zero measurements, which corresponds to the two measurement patterns for an ancilla qubit, $\{0, 1\}$, in the previous studies [18, 13]. Then, as shown just later, the likelihood function with respect to ε can be well approximated by the Gaussian $P(\vec{0}|\varepsilon) = \mathcal{N}(\varepsilon; \mu_{\text{lh}}, \sigma^2_{\text{lh}})$. As a result, the posterior distribution keeps the form of Gaussian as $P(\varepsilon|\vec{0}) = \mathcal{N}(\varepsilon; \mu_{\text{post}}, \sigma^2_{\text{post}})$, where

$$\mu_{\text{post}} = \frac{\sigma^2_{\text{prior}}\mu_{\text{lh}} + \sigma^2_{\text{lh}}\mu_{\text{prior}}}{\sigma^2_{\text{lh}} + \sigma^2_{\text{prior}}}, \tag{S1.2}$$

$$\sigma^2_{\text{post}} = \frac{\sigma^2_{\text{prior}}\sigma^2_{\text{lh}}}{\sigma^2_{\text{lh}} + \sigma^2_{\text{prior}}}. \tag{S1.3}$$

In the circuit implementation in Fig. 1(c), we assume

$$U_{\text{prep}} |0\rangle^{\otimes N+1} \approx \frac{1}{\sqrt{2}}(|0\rangle |\psi_{\text{g}}\rangle + |1\rangle |\psi_{\text{ex}}\rangle), \tag{S1.4}$$

$$U_{\text{evol}} \approx e^{-iH\Delta t}, \tag{S1.5}$$

where $|\psi_{\text{g}}\rangle$ and $|\psi_{\text{ex}}\rangle$ are the ground and excited states, respectively, and N is the number of system qubits. The wave function after operating $U^{t/\Delta t}_{\text{evol}}$ and $P(\varepsilon t)$ is

$$\begin{aligned} &(P(\varepsilon t) \otimes U^{t/\Delta t}_{\text{evol}}) \frac{1}{\sqrt{2}}(|0\rangle |\psi_{\text{g}}\rangle + |1\rangle |\psi_{\text{ex}}\rangle) \\ &\approx \frac{1}{\sqrt{2}}(e^{-iE_{\text{g}}t} |0\rangle |\psi_{\text{g}}\rangle + e^{-i(E_{\text{ex}}-\varepsilon)t} |1\rangle |\psi_{\text{ex}}\rangle), \end{aligned} \tag{S1.6}$$

where E_{g} and E_{ex} are the eigenvalues of the ground and excited states, respectively. Suppose here the state before the measurement as

$$|\phi\rangle = \frac{1}{\sqrt{2}} U^{\dagger}_{\text{prep}}(e^{-iE_{\text{g}}t} |0\rangle |\psi_{\text{g}}\rangle + e^{-i(E_{\text{ex}}-\varepsilon)t} |1\rangle |\psi_{\text{ex}}\rangle). \tag{S1.7}$$

Algorithm 1 Tensor-based quantum phase difference estimation

Input: Initial mean (variance) as a prior Gaussian distribution μ_{init} (σ^2_{init}), the number of sample points m, the number of shots R, single time step Δt

▷ Initial preparation

$\mu_{\text{prior}} \leftarrow \mu_{\text{init}}$
$\sigma^2_{\text{prior}} \leftarrow \sigma^2_{\text{init}}$
$iter \leftarrow 1$

▷ Estimation loop

while true **do**
    $t \leftarrow 1.8/\sigma^2_{\text{prior}}$
    $\vec{\varepsilon}_{\text{list}} \leftarrow$ make a list by selecting m points equally in the interval from $\mu_{\text{prior}} - \sigma^2_{\text{prior}}$ to $\mu_{\text{prior}} + \sigma^2_{\text{prior}}$
    $\vec{p}_{\text{list}} \leftarrow$ empty list
    for all ε **in** $\vec{\varepsilon}_{\text{list}}$ **do**
        Execute the circuit in Fig. 1(c) with t, Δt, ε, and shot number R
        Obtain the probability p that 0 is measured in all qubits of the circuit
        Append the value pair (ε, p) to $\vec{p}_{\text{list}}$
    end for

    ▷ Mean and variance of likelihood function

    $\mu_{\text{lh}}, \sigma^2_{\text{lh}} \leftarrow$ calculate from the Gaussian fitting of $\vec{p}_{\text{list}}$

    ▷ Mean and variance of posterior distribution

    Calculate $\mu_{\text{post}}, \sigma^2_{\text{post}}$ by Eqs. (S1.2) and (S1.3) using $\mu_{\text{prior}}, \sigma^2_{\text{prior}}, \mu_{\text{lh}}$, and σ^2_{lh}
    Print $iter, \vec{p}_{\text{list}}, \mu_{\text{post}}$, and σ^2_{post}
    if a termination condition, such as threshold of σ^2_{post}, is satisfied **then**
        break
    else
        $\mu_{\text{prior}} \leftarrow \mu_{\text{post}}$
        $\sigma^2_{\text{prior}} \leftarrow \sigma^2_{\text{post}}$
        $iter \leftarrow iter + 1$
    end if
end while

Then, the probability of having the measurement result $\vec{x} = \vec{0}$, i.e., the likelihood function $P(\vec{0}|\varepsilon)$, is given by

$$\begin{aligned} P(\vec{0}|\varepsilon) &= \mathrm{Tr}\left[(|0\rangle\langle 0|)^{\otimes N+1} |\phi\rangle\langle\phi|\right] \\ &\approx \frac{1}{4}|(\langle 0| \langle\psi_{\mathrm{g}}| + \langle 1| \langle\psi_{\mathrm{ex}}|)(e^{-iE_{\mathrm{g}}t} |0\rangle |\psi_{\mathrm{g}}\rangle + e^{-i(E_{\mathrm{ex}}-\varepsilon)t} |1\rangle |\psi_{\mathrm{ex}}\rangle)|^2 \\ &= \frac{1}{4}|e^{-iE_{\mathrm{g}}t} + e^{-i(E_{\mathrm{ex}}-\varepsilon)t}|^2 \\ &= \frac{1}{2}(1 + \cos(E_{\mathrm{gap}} - \varepsilon)t), \end{aligned} \tag{S1.8}$$

and the final expression can be approximated as

$$P(\vec{0}|\varepsilon) \approx \exp\left\{-\frac{t^2}{4}\left(\varepsilon - E_{\mathrm{gap}}\right)^2\right\}, \tag{S1.9}$$

where E_{gap} is a energy gap represented as $E_{\mathrm{gap}} = E_{\mathrm{ex}} - E_{\mathrm{g}}$. Therefore, by calculating the probability that all qubits are measured at 0 in an appropriate range of ε, the likelihood function can be approximated by a Gaussian distribution.

The number of two-qubit gates required to execute the circuit of Fig. 1(c) is at most

$$3 \times \left\{ N \times \lceil \frac{d_{\mathrm{prep}}}{2} \rceil \times 2 + (N-1) \times \lceil \frac{d_{\mathrm{evol}}}{2} \rceil \times \frac{t}{\Delta t} \right\}, \tag{S1.10}$$

where the first 3 comes from the fact that any two-qubit gate can be executed with up to 3 native two-qubit gates [16, 2]. In the real device execution, the gate count would be smaller than the above value due to the transpilation and the error suppression, where optimization level 3 was adopted in the Qiskit transpilation.

Our phase difference estimation algorithm is implemented on Qiskit [8] with the FireOpal error suppression module of Q-CTRL [10, 12] by the Python language. FireOpal is a package of deterministic error-suppression workflow, and the overview of the execution procedure is as follows [10]; a user first create quantum circuits (as a quantum assembly language, QASM) and submit them to Q-CTRL through the FireOpal module. The module first executes the front-end compiler including mathematical reduction of the gate count of quantum circuits. Then it executes an error-reducing back-end compiler including an error-aware hardware mapping, an elimination of circuit crosstalk (dynamical decoupling), and an optimized gate replacement (e.g., AI-powered analog-level gate optimization). The created circuits are experimented on a real device. Before returning the results to the user, the module performs a measurement error mitigation including an AI-driven calibration routine.

We choose $m = 21$, $R = 10,000$, and $\Delta t = 0.1$. σ^2_{init} is 9.0 in the result of Fig. 2 and is 4.0 in the other results. $\mu_{\mathrm{init}} = 0.0$ in most of the cases, but when the peak of the likelihood function diverged in the noiseless simulation, it deviated by 0.01. In addition, in the molecular orbital model of hexatriene, the

lack of precision in the approximation of U_{evol} sometimes caused the Gaussian fit to fail because the peak position deviated significantly from the value of the previous iteration. In such cases, the value of ε which has the maximum p was assigned to μ_{prior} and restarted. The total time t is chosen as $1.8/\sigma^2_{\mathrm{prior}}$. For the noiseless simulation, the termination condition of our algorithm is achieving $\sigma^2_{\mathrm{post}} \leq 0.005$. In the real device execution, the algorithm stopped around 4,000-8,000 two-qubit gates in a circuit before the suppression due to FireOpal (or IBM Quantum Platform) system errors, e.g., QASM upload size limit, before reaching the above conditions excluding octatetraene in Supporting Information S5, where we accidentally reached the termination condition in octatetraene because of noise contamination. We also stopped iterations when the Gaussian peak was 0 or the Gaussian fit failed due to small signals and noise affections. The prior Gaussian distribution is used as the initial guess of the Gaussian fitting of the likelihood function.

S2 Unitary compression

We approximate the time evolution in a single step $e^{-iH\Delta t}$ by U_{evol}, where the error in the simulation for a total time t, $\left\|e^{-iHt} - U_{\mathrm{evol}}^{\frac{t}{\Delta t}}\right\|_F$, is bounded as

$$\begin{aligned}
\left\|e^{-iHt} - U_{\mathrm{evol}}^{\frac{t}{\Delta t}}\right\|_F &= \left\|\sum_{k=1}^{t/\Delta t} e^{-iH(\frac{t}{\Delta t}-k)}(e^{-iH\Delta t} - U_{\mathrm{evol}})U_{\mathrm{evol}}^{k-1}\right\|_F \\
&\leq \sum_{k=1}^{t/\Delta t} \left\|e^{-iH(\frac{t}{\Delta t}-k)}(e^{-iH\Delta t} - U_{\mathrm{evol}})U_{\mathrm{evol}}^{k-1}\right\|_F \\
&\leq \sum_{k=1}^{t/\Delta t} \left\|e^{-iH(\frac{t}{\Delta t}-k)}\right\| \left\|e^{-iH\Delta t} - U_{\mathrm{evol}}\right\|_F \left\|U_{\mathrm{evol}}^{k-1}\right\| \\
&= \frac{t}{\Delta t}\left\|e^{-iH\Delta t} - U_{\mathrm{evol}}\right\|_F,
\end{aligned} \tag{S2.1}$$

$\|\cdot\|_F$ is the Frobenius norm, and $\|\cdot\|$ is the operator norm. We used equality $A^n - B^n = \sum_{k=1}^{n} A^{n-k}(A-B)B^{k-1}$ in the second expression where A and B are matrices, and n is an integer, triangle inequality in the third expression, and the inequalities $\|AB\|_F \leq \|A\|\|B\|_F$ and $\|BA\|_F \leq \|B\|_F\|A\|$ in the fourth expression. The last inequalities can be shown as $\|AB\|_F^2 = \mathrm{Tr}\big((AB)^\dagger AB\big) = \mathrm{Tr}\big(B^\dagger A^\dagger AB\big) \leq \lambda_{max}(A^\dagger A)\,\mathrm{Tr}\big(B^\dagger B\big) = \|A\|^2\|B\|_F^2$ where $\lambda_{max}(A^\dagger A)$ is the maximum eigenvalue of $A^\dagger A$, corresponding to the squared operator norm (and the squared maximum singular value) of A, and $A^\dagger A \leq \lambda_{max}(A^\dagger A) I^{\otimes N}$ [19]. $\|BA\|_F \leq \|B\|_F\|A\|$ is also shown in almost the same procedure.

The unitary compression of U_{evol} is based on the procedure in Ref. [1]. We first prepare an MPO $U_{\mathrm{ref}}^\dagger$ corresponding to $e^{iH\Delta t}$ by using the second-order Trotter approximation for 100 sliced time steps, $\Delta t/100$. Specifically, we start from an identity MPO and multiply by the time evolution MPO of the sliced

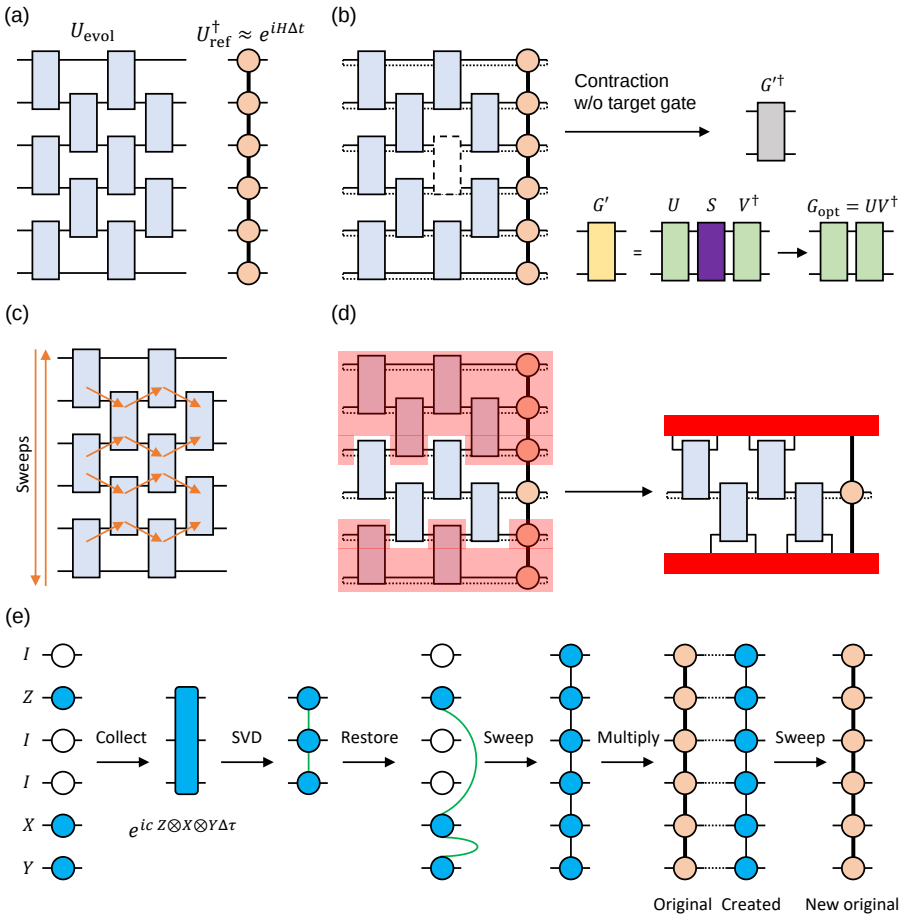


Figure S1: Procedure of a unitary compression. Here is an example of six qubits and $d_{\mathrm{evol}} = 4$. (a) Introduction of MPOs. Horizontal lines represent qubits. Vertical bold lines in $U_{\mathrm{ref}}^{\dagger}$ represent MPO virtual bonds. (b) Gate optimization. The dashed box represents a target gate to optimize. The dotted line connecting the left end and the right end represents a trace. (c) Optimization sequence. (d) Environmental tensors. (e) Example of creation of time evolution operator of $\exp\{ic(I \otimes Z \otimes I \otimes I \otimes X \otimes Y)\Delta\tau\}$.

step for each term of H, which is expanded by the tensor product of Pauli matrices. To prevent divergence of the bond dimension, a sweep of singular value decomposition (SVD) truncating singular values below a threshold value of 10^{-12} was performed for the MPO each time applying the time evolution of the term. Next, we optimize a brick-wall MPO U_{evol} to approximate U_{ref}. We

consider minimizing the squared Frobenius norm of these operators,

$$\begin{aligned}\|U_{\mathrm{ref}} - U_{\mathrm{evol}}\|_F^2 &= \mathrm{Tr}\Big[U_{\mathrm{ref}}^\dagger U_{\mathrm{ref}}\Big] + \mathrm{Tr}\Big[U_{\mathrm{evol}}^\dagger U_{\mathrm{evol}}\Big] - 2\,\mathrm{Re}\,\mathrm{Tr}\Big[U_{\mathrm{ref}}^\dagger U_{\mathrm{evol}}\Big] \\ &= 2^{N+1} - 2\,\mathrm{Re}\,\mathrm{Tr}\Big[U_{\mathrm{ref}}^\dagger U_{\mathrm{evol}}\Big].\end{aligned} \tag{S2.2}$$

The norm can be minimized by maximizing the $\mathrm{Re}\,\mathrm{Tr}\Big[U_{\mathrm{ref}}^\dagger U_{\mathrm{evol}}\Big]$ of the last term.

Figure S1 shows the optimization procedure. The orange tensor network represent $U_{\mathrm{ref}}^\dagger$, and the blue brick-wall tensor network represent U_{evol} in Fig. S1(a). We explain how to optimize each of the gates in U_{evol}; $\mathrm{Re}\,\mathrm{Tr}\Big[U_{\mathrm{ref}}^\dagger U_{\mathrm{evol}}\Big]$ can be transformed as $\mathrm{Re}\,\mathrm{Tr}\Big[(U_2 U_{\mathrm{ref}}^\dagger U_1)G\Big]$ by representing $U_{\mathrm{evol}} = U_1 G U_2$ using the target gate G and the the other gates U_1 and U_2. Then we consider finding the optimal G. $\mathrm{Re}\,\mathrm{Tr}\Big[(U_2 U_{\mathrm{ref}}^\dagger U_1)G\Big]$ can be represented as $\mathrm{Re}\,\mathrm{Tr}\big[S(U^\dagger G V)\big]$ using the SVD $U_2 U_{\mathrm{ref}}^\dagger U_1 = VSU^\dagger$, where U and V are unitary matrices, and S is a singular value (diagonal) matrix. $U^\dagger G V$ can be regarded as a unitary matrix, that is, each of the columns is a normalized vector, and thus $\mathrm{Re}\,\mathrm{Tr}\big[S(U^\dagger G V)\big] \leq \mathrm{Re}\,\mathrm{Tr}[S]$. Therefore, when $U^\dagger G V$ is identity, i.e., $G = UV^\dagger$, $\mathrm{Re}\,\mathrm{Tr}\Big[U_{\mathrm{ref}}^\dagger U_{\mathrm{evol}}\Big]$ takes a maximum value.

To get the optimal gate in the actual implementation, we first contract all of the tensors except for the target gate and obtain a four-leg tensor $G'^\dagger$ in the left and right top panels of Fig. S1(b). As in the right bottom panel of Fig. S1(b), the SVD is executed for G', i.e., $G' = USV^\dagger$. Then, we obtain the optimal gate $G_{\mathrm{opt}} = UV^\dagger$ by replacing S to identity. This optimization is performed for each gate. As shown in Fig. S1(c), gate optimization is performed in the order of sweeping up and down the zigzag path from left to right. The initial gate was prepared by QR decomposition of the identity operator plus a small random unitary operator. The number of sweeps was 1,000 for the Hubbard models and 10,000 for linear polyenes. To reduce the number of contractions, the upper and lower portions of the target gate are stored as an environmental tensor (red) and used as needed as in Fig. S1(d), and the cost of contracting the tensor network is $\mathcal{O}\big(\chi d_{\mathrm{evol}} 2^{4+2\lceil d_{\mathrm{evol}}/2\rceil} + \chi^2 2^{2+2\lceil d_{\mathrm{evol}}/2\rceil}\big)$, where χ is the maximum bond dimension in $U_{\mathrm{ref}}^\dagger$ [1]. Note that in this study, when creating the $U_{\mathrm{ref}}^\dagger$ in Fig. S1(a), we reduce the computational cost by calculating the time evolution operators by excluding the identity operators from the tensor product of Pauli operators in a Hamiltonian. Specifically, as in the example of Fig. S1(e), the following procedure is repeated for each term of the Hamiltonian; collecting only the Pauli X, Y, and Z operators, performing time evolution operations, restoring them to their initial MPO sites using SVD, converting them to MPO by sweeping, and multiplying the created MPO by the original MPO to make a new original MPO. Thus, the cost of SVD is exponentially dependent on the locality of a Pauli term, rather than N. In addition, this exponential increase could be avoided by decomposing the collected exponential operator into local

controlled-NOT and one-qubit gates. The initial original MPO is identity, c is a coefficient in the term, and $\Delta\tau = \Delta t/200$ in our case of second-order Trotter approximation.

To execute the unitary compression of $U_{\rm prep}$, we need to prepare $|{\rm MPS}\rangle = \frac{1}{\sqrt{2}}(|0\rangle|\psi_{\rm g}\rangle + |1\rangle|\psi_{\rm ex}\rangle)$. We calculate $|\psi_{\rm g}\rangle$ by the DMRG, and $|\psi_{\rm g}\rangle$ is an MPS as

$$|\psi_{\rm g}\rangle = \sum_{\vec{a}}\sum_{\vec{s}} A_{s_1 a_1} A_{s_2 a_1 a_2} \dots A_{s_N a_{N-1}} |\vec{s}\rangle\,, \tag{S2.3}$$

where $\vec{a} = a_1 a_2 \dots a_{N-1}$, $a_\iota \in \{0, 1, \dots, \chi_\iota - 1\}$ $(\iota = 1, 2, \dots, N-1)$ represents a virtual index, χ_ι is a bond dimension, $\vec{s} = s_1 s_2 \dots s_N$, and $s_\kappa \in \{0, 1\}$ $(\kappa = 1, 2, \dots, N)$ represents a physical index. $|0\rangle|\psi_{\rm g}\rangle$ can be represented as an MPS using a dummy tensor A_{s_0},

$$|0\rangle|\psi_{\rm g}\rangle = \sum_{\vec{a}}\sum_{s_0,\vec{s}} A_{s_0} A_{s_1 a_1} A_{s_2 a_1 a_2} \dots A_{s_N a_{N-1}} |s_0\vec{s}\rangle\,, \tag{S2.4}$$

where A_{s_0} $(s_0 \in \{0, 1\})$ is a tensor of $A_0 = 1$ and $A_1 = 0$. $|0\rangle|\psi_{\rm ex}\rangle$ can also be expressed in an MPS by the same procedure as above,

$$|1\rangle|\psi_{\rm ex}\rangle = \sum_{\vec{a}'}\sum_{s_0,\vec{s}} A'_{s_0} A'_{s_1 a'_1} A'_{s_2 a'_1 a'_2} \dots A'_{s_N a'_{N-1}} |s_0\vec{s}\rangle\,, \tag{S2.5}$$

where $\vec{a}'$ is denoted in the same way as $\vec{a}$ previously mentioned, and A'_{s_0} is a tensor of $A'_0 = 0$ and $A'_1 = 1$. We add the two states to prepare an unnormalized superposition state,

$$|0\rangle|\psi_{\rm g}\rangle + |1\rangle|\psi_{\rm ex}\rangle = \sum_{\tilde{a}_0\vec{\tilde{a}}}\sum_{s_0,\vec{s}} \tilde{A}_{s_0\tilde{a}_0} \tilde{A}_{s_1\tilde{a}_0\tilde{a}_1} \tilde{A}_{s_2\tilde{a}_1\tilde{a}_2} \dots \tilde{A}_{s_N\tilde{a}_{N-1}} |s_0\vec{s}\rangle\,, \tag{S2.6}$$

where

$$\begin{aligned}
\tilde{A}_{s_0\tilde{a}_0} &= \begin{cases} A_{s_0} & \text{if } \tilde{a}_0 = a_0 \\ A'_{s_0} & \text{if } \tilde{a}_0 = a'_0 \end{cases} \\
\tilde{A}_{s_1\tilde{a}_{\iota-1}\tilde{a}_\iota} &= \begin{cases} A_{s_\iota a_{\iota-1} a_\iota} & \text{if } \tilde{a}_{\iota-1} = a_{\iota-1}, \tilde{a}_\iota = a_\iota \\ A'_{s_\iota a'_{\iota-1} a'_\iota} & \text{if } \tilde{a}_{\iota-1} = a'_{\iota-1}, \tilde{a}_\iota = a'_\iota \\ 0 & \text{otherwise} \end{cases} \\
\tilde{A}_{s_N\tilde{a}_{N-1}} &= \begin{cases} A_{s_N a_{N-1}} & \text{if } \tilde{a}_{N-1} = a_{N-1} \\ A'_{s_N a'_{N-1}} & \text{if } \tilde{a}_{N-1} = a'_{N-1} \end{cases}
\end{aligned} \tag{S2.7}$$

$\vec{\tilde{a}} \in \{\vec{a}, \vec{a}'\}$, and $\tilde{a}_0 \in \{a_0 = 0, a'_0 = 1\}$. That is, the superposition state can be represented as an MPS. $|{\rm MPS}\rangle$ is obtained by normalization and transformation to the left canonical form of the superposition state.

The optimization procedure of a brick-wall MPO $U_{\rm prep}$ to approximate $|{\rm MPS}\rangle$ is almost the same as $U_{\rm evol}$ since we can execute the same procedure by substituting $U_{\rm evol}$ to $U_{\rm prep}$ and $U_{\rm ref}^\dagger$ to $|0\rangle^{\otimes N+1}\langle{\rm MPS}|$ in $\mathrm{Re\,Tr}\left[U_{\rm ref}^\dagger U_{\rm evol}\right]$ of Eq. (S2.2), that is,

$$\mathrm{Re\,Tr}\left[|0\rangle^{\otimes N+1}\langle{\rm MPS}|\,U_{\rm prep}\right] = \langle{\rm MPS}|\,U_{\rm prep}\,|0\rangle^{\otimes N+1}\,, \tag{S2.8}$$

which is the same as in Eq. (2). The number of sweeps was 1,000 for all the models. Note that MPS can be embedded with sequential gates without optimization [11], but it was not adopted in our case because of the increasing depth and idle time of each qubit.

The remaining calculation conditions are described as follows. The unitary compression was implemented in ITensor [3] package of the Julia language. Hamiltonian prepared by Qiskit (see Sec. S3) was converted to an ITensor format, and U_{prep} and U_{evol} obtained by unitary compressions were saved in NumPy [7] format, and then the phase difference estimation was performed using Qiskit with the compressed gates. The number of the DMRG sweeps when preparing $|\mathrm{MPS}\rangle$ was set to 20, the maximum bond dimension in SVD was set to 10 for the first 3 sweeps, 50 for the next 12 sweeps, and 1,000 for the last 5 sweeps, and the cutoff threshold in SVD was 10^{-12} for the Hubbard models and 10^{-8} for linear polyenes. The DMRG of the excited states was executed to reduce overlap with the MPS of the ground state. Note that in ITensor's code implementation, the MPS of Eq. (S2.6) is obtained simply by adding Eq. (S2.4) and Eq. (S2.5), i.e., and we did not need to construct the tensor of Eq. (S2.6) explicitly.

S3 Model construction

Qiskit [8] was used for the fermionic model construction and Jordan–Wigner transformation for the Hubbard models and linear polyenes. The electronic structures of linear polyenes were calculated by the PySCF package [14], and the geometry were optimized using the Hartree–Fock method with the 6-31G* basis set by the PySCF and geomeTRIC [17] packages. The molecular orbitals were localized by Boys localization [5] for π and π^* orbitals. Fermionic Hamiltonians for hexatriene, octatetraene and decapentaene were constructed as 12-qubit CASCI (6e 6o), 16-qubit CASCI (8e 8o), and 20-qubit CASCI (10e 10o) problems, respectively.

The orbital order was optimized using a genetic algorithm implemented with the DEAP library [4]. The cost function is C in Eq. (4), the parameters were set with a population size of 50, a crossover probability of 70%, a mutation probability of 20%, and 100 generations. Ordered crossover and index-shuffling mutation were applied, and tournament selection with a size of three was used. Note that while we utilized the Jordan–Wigner transformation after the reordering, incorporating fermion-qubit mappings considering circuit topology [15, 9] are left for future work.

S4 Time scale for the quantum advantage

The maximum time length taken in this study seems to be within the range where classical computers can simulate using the MPO framework. To estimate a classically-simulable regime, we additionally studied the time dependence of

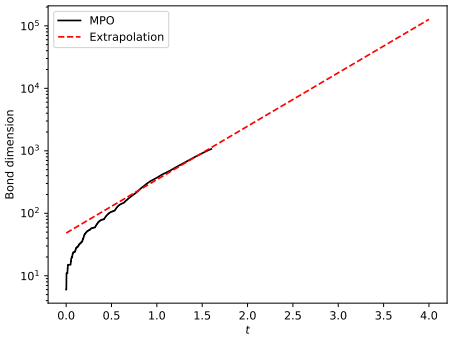


Figure S2: Bond dimension of the time evolved MPO versus the time duration t.

the bond dimension of MPO for the Hubbard model with 20 system qubits. The results are shown in Fig. S2. The MPO sliced time step width is the same as in this study, $\Delta t/100 = 10^{-3}$. We calculate it the bond dimension up to $t = 1.6$ (black curve) and extrapolate the results by fitting it to an exponential function (red line). Considering that the limit of the bond dimension that can be executed in HPC is 10^4-10^5 [6], the classical calculation would become difficult in the timescale longer than $t \approx 3.5$. In our demonstration, 20 qubit system real device execution was able to run up to $t = 3.3$, so it appears that this naive comparison shows comparable performance. Therefore, a quantum advantage can be expected in the near future with improved device performance and algorithms. We finally note that by taking into account the tight cutoff threshold of SVD 10^{-12} and the approximation errors in the transformation from the MPO to a brick-wall circuit, minor modifications would lean the advantage to the classical side in the current situation.

S5 Error analysis for the algorithm

We note the error robustness in our algorithm. Assume the global depolarization channel for the state before the measurement $|\phi\rangle$ in Eq. (S1.7),

$$\mathcal{E}(|\phi\rangle\langle\phi|) = (1 - p_{\mathrm{dep}})\,|\phi\rangle\langle\phi| + p_{\mathrm{dep}}\frac{1}{2^{N+1}}I^{\otimes N+1}, \tag{S5.1}$$

where p_{dep} is an error rate of the depolarization channel. The probability in Eq. (S1.8) becomes

$$\mathrm{Tr}\left[(|0\rangle\langle 0|)^{\otimes N+1}\mathcal{E}(|\phi\rangle\langle\phi|)\right] \approx (1 - p_{\mathrm{dep}})\frac{1}{2}(1 + \cos(E_{\mathrm{gap}} - \varepsilon)t) + \frac{p_{\mathrm{dep}}}{2^{N+1}}, \tag{S5.2}$$

and the error does not change the peak position while the amplitude decreases. In addition, the second term, which comes from the effect of depolarization noise, is exponentially decaying with respect to N. When calculated with a

similar setup in QPE [Fig. 1(a)] and QPDE [Fig. 1(b)], the second term becomes $p_{\mathrm{dep}}/2$, i.e., constant, due to the ancilla-only measurement [18]. Therefore, the exponentially small noise affection is a feature of our algorithm.

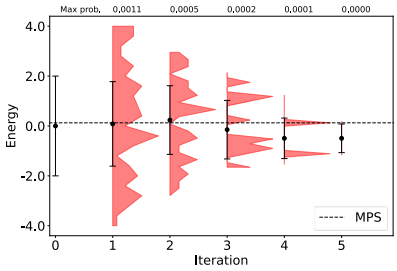


Figure S3: Demonstration of our algorithm in the one-dimensional Hubbard models using the Heron device *ibm_torino* without the error suppression (21-qubit circuit). The circle and error bar denote the mean value and standard deviation, respectively, of the posterior distribution in the iteration. The sampled probabilities are shown in red, and each plot is normalized by the maximum value in each iteration, which is shown at the top of the plot.

The affection of the decay actually can be seen in the results of the real device execution. For example, for the eight-qubit Hubbard model, the value of the second term of noise is $2^{-9} \approx 2.0 \times 10^{-3}$ at maximum. In fact, a uniform distribution of 0.001 magnitudes can be seen in the latter half of the iterations in Fig. 3(b). On the other hand, the values in 21- and 33-qubit circuits are $2^{-21} \approx 4.8 \times 10^{-7}$ and $2^{-33} \approx 1.2 \times 10^{-10}$, respectively, and the sharp peak can be seen of Fig. 3(d) despite a small signal of 0.001–0.01 magnitude. As related data, the 20 qubit Hubbard model without error suppression is shown in Fig. S3. A small peak is seen at the beginning, but the signal disappears with the proceeding iteration. In the final iteration, there is no signal at all, meaning that there is not only no signal but also no noise effect. The same tendency was found in the octatetraene (17-qubit circuit) in Fig. S4 (a) and (c); only a few shots were detected in the final iteration since the number of detection from the noise affection is estimated as $2^{-17} \times 10,000 \times 21 \approx 1.6$ under 10,000 shots/circuit and 21 circuits/iteration, which led to a coincidentally sudden convergence in the final iteration. Here, $d_{\mathrm{prep}} = d_{\mathrm{evol}} = 10$, the gap in the iteration of the final (before the final) is 0.525 ± 0.051 (0.048 ± 1.109), the exact gap is 0.113, and the number of two-qubit gates is 4056/638 (1794/504). We also executed decapentaene (21 qubit-circuit) in Fig. S4 (b) and (d), and the gap is 0.009 ± 1.005, where $d_{\mathrm{prep}} = 8, d_{\mathrm{evol}} = 10$, the exact gap is 0.105, and a two-qubit gate count is 2730/693. These results suggest that our algorithm can estimate the energy for large qubits as long as the signal is detected.

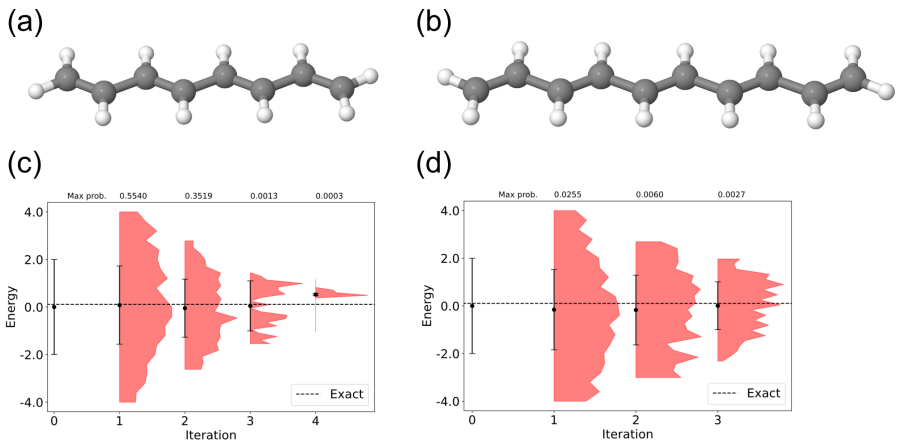


Figure S4: Demonstration of our algorithm in octatetraene (17-qubit circuit) and decapentaene (21-qubit circuit) using the Heron devices *ibm_torino* and *ibm_fez*, respectively, with the error suppression. (a) the structure of octatetraene. (b) the structure of decapentaene. (c) the demonstration of our algorithm of octatetraene. The circle and error bar denote the mean value and standard deviation, respectively, of the posterior distribution in the iteration. The sampled probabilities are shown in red, and each plot is normalized by the maximum value in each iteration, which is shown at the top of the plot. (d) the demonstration of our algorithm of decapentaene.

S6 FCI energy calculation

The FCI energy can be calculated by substituting $|\psi_{\mathrm{ex}}\rangle$ for a vacuum state $|0\rangle^{\otimes N}$ in the superposition state. Since the vacuum state is trivially MPS and the energy of the vacuum state is zero, Eq. (S1.8) becomes

$$\mathrm{Tr}\left[(|0\rangle\langle 0|)^{\otimes N+1}|\phi\rangle\langle\phi|\right] \approx \frac{1}{2}(1+\cos(E_{\mathrm{g}}+\varepsilon)t), \tag{S6.1}$$

the (sign reversed) ground state energy can be calculated in our algorithm. Fig. S5 shows the demonstration of the eight-qubit Hubbard model, where U_{evol} was shared with the gap estimation in the main text, U_{prep} was prepared by $d_{\mathrm{prep}} = 6$ with $f(U_{\mathrm{prep}};\ |\mathrm{MPS}\rangle) = 0.99$, and μ_{init} was set to the value when the DMRG is performed with the maximum bond dimension as two. The results of the noiseless simulator in Fig. S5(a) show that the distribution approaches the exact solution with each iteration. The final value is -20.866 ± 0.005, where the exact solution is -20.911. We also confirmed that the algorithm works on the real device (especially when the maximum probability in an iteration is greater than 0.01) in Fig. S5(b), where the energy of a final iteration is -21.326 ± 0.444.

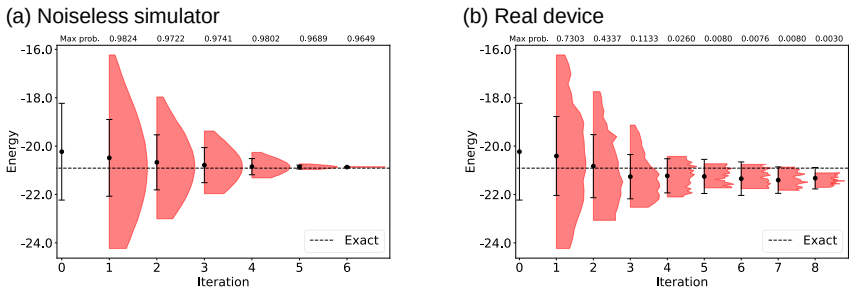


Figure S5: Demonstration of our algorithm for the FCI calculation in the Hubbard model (nine-qubit circuit) using (a) the noiseless simulator and (b) the Heron device *ibm_torino* with the error suppression. For ease of understanding, the signs of energy value are reversed. The circle and error bar denote the mean value and standard deviation, respectively, of the posterior distribution in the iteration. The sampled probabilities are shown in red, and each plot is normalized by the maximum value in each iteration, which is shown at the top of the plot.

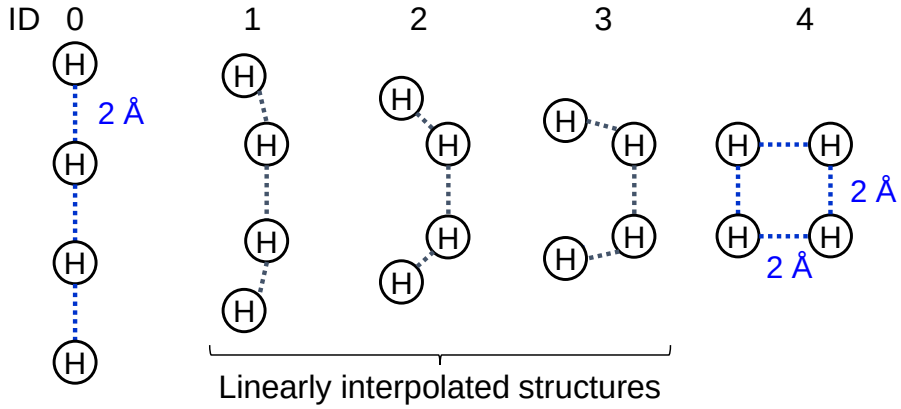


Figure S6: Calculated hydrogen cluster models. The bond lengths between two hydrogen atoms in ID 0 and 4 are 2 Angstrom. The three structures are linearly interpolated between ID 0 and 4.

S7 Additional benchmarks for interaction complexity

We performed our algorithm on five structures of H_4 clusters that gradually changed from linear to square structures using noiseless simulations. We show the result in Table S1, where $d_{\mathrm{prep}} = d_{\mathrm{evol}} = 8$. We found that all of the models achieved a level of accuracy for the error similar to that of the hexatriene. How-

Table S1: Results of hydrogen cluster models. The energy unit is Hartree. "Gap (TQPDE)" and "Gap (exact)" are the energy gaps of the tensor-based QPDE and exact, respectively. "Error" is the difference between Gap (TQPDE) and Gap (exact).

ID	$f(U_{\text{prep}};\ \lvert\text{MPS}\rangle)$	$\delta(U_{\text{evol}};\ U_{\text{ref}})$	Gap (TQPDE)	Gap (exact)	Error
0	1.00	0.0055	0.122	0.0159	0.1057
1	1.00	0.0068	0.200	0.0793	0.1212
2	1.00	0.0074	0.276	0.1283	0.1481
3	1.00	0.0076	0.218	0.0769	0.1406
4	0.92	0.0137	-0.028	0.0171	0.0448

ever, from the metrics of MPS and MPO, $f(U_{\text{prep}};\ |\text{MPS}\rangle)$ and $\delta(U_{\text{evol}};\ U_{\text{ref}})$, respectively, we found that as the interaction changed from one-dimensional to two-dimensional, the accuracy tended to decrease. Therefore, although our approach can be applied to complex interactions, it was found to be more effective for systems that are close to one-dimensional.

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